Simulated annealing floorplan search

Toy cost adds one thousand when a packing is illegal

Pseudocode

- Floorplan annealing uses a cost that heavily penalizes illegality
- Pseudocode proposes neighbors, accepts by Metropolis, and keeps the best legal iterate
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Bad overflow seed costs about one thousand forty-four
- Golden legal packing is about thirty-six
- One teaching improve step replaces bad with golden

Algorithm sketch — try these

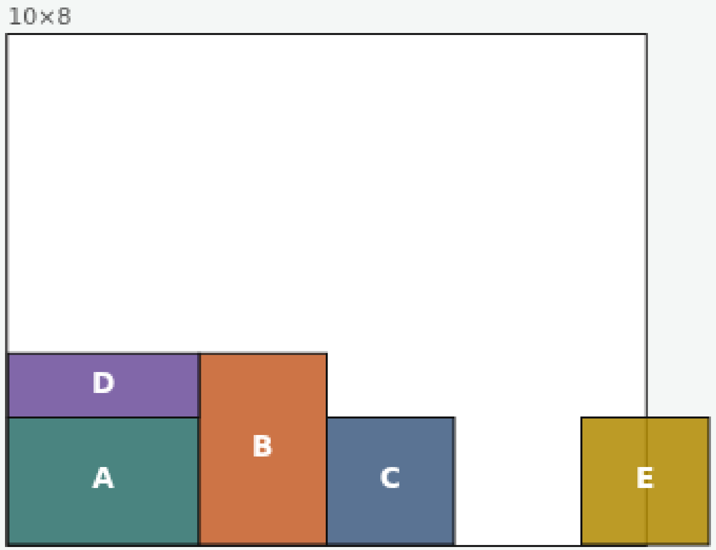
```
INPUT: pack / representation, T schedule
OUTPUT: best legal low-cost pack
cost ← 1000·¬legal + deadspace + α·HPWL
propose neighbor (swap/move/perturb)
accept if Δ<0 or rand < e^(−Δ/T)
keep best; cool T
GOLDEN legal cost≈36; bad≈1044
```

Illegal packs pay 1000

SIMULATED ANNEALING FP

Illegal packs pay 1000

Step 1 / 5 · bad-cost · module03-01-simulated-annealing-fp



Explanation

Our toy cost adds one thousand when the packing is illegal. The bad overflow seed therefore sits at cost about one thousand forty-four—dominated by the penalty.

- illegal → +1000
- Plus deadspace and HPWL terms
- Never accept illegal as “good”

```
cost(bad)≈1044.4
legal: false
```

Illegal packs pay 1000

Golden cost stays under 1000

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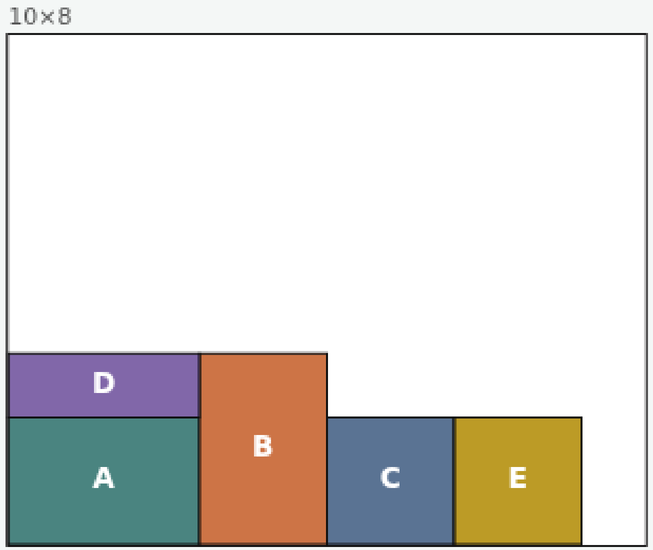
Golden cost stays under 1000

Step 2 / 5 · golden-cost · module03-01-simulated-annealing-fp



Explanation

The golden legal packing drops below the penalty floor. Cost is about thirty-six—deadspace and a small HPWL proxy, no illegality tax.

- cost(golden)≈35.7
- legal: true
- Beats bad by ~1000

```
cost(golden)≈35.7
hpwl≈12.5
```

Golden cost stays under 1000

Neighbors swap module positions

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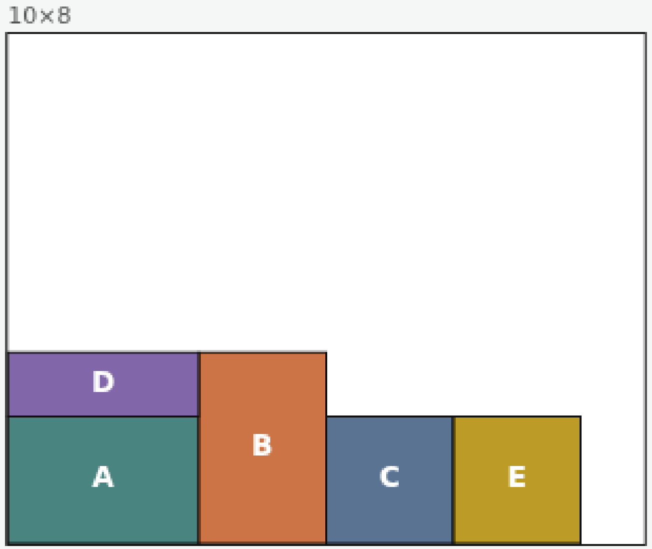
Neighbors swap module positions

Step 3 / 5 · neighbor · module03-01-simulated-annealing-fp



Explanation

A simple SA move swaps the lower-left corners of two modules while keeping sizes. Accept improving moves; accept worsening ones with temperature probability.

- saSwap(A,E) exchanges coordinates
- Sizes unchanged
- Re-check legality after the move

```
move: coordinate swap
then rescore cost
```

Neighbors swap module positions

Improve: bad → golden

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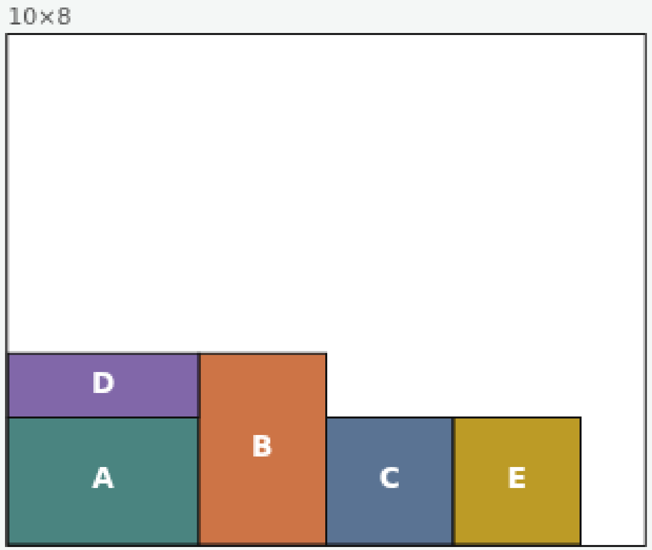
Improve: bad → golden

Step 4 / 5 · improve · module03-01-simulated-annealing-fp



Explanation

One teaching “improve” step replaces the illegal seed with the golden packing. Cost falls below one thousand and legality flips to true—exactly what a cooling schedule should prefer.

- cost drops below 1000
- legal becomes true
- HPWL becomes finite and meaningful

```
before≈1044.4
after≈35.7
```

Improve: bad → golden

SA needs a representation

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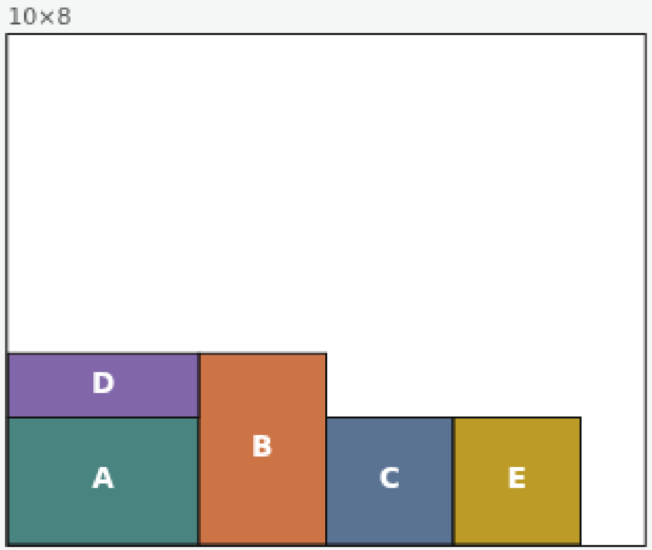
SA needs a representation

Step 5 / 5 · takeaway · module03-01-simulated-annealing-fp



Explanation

Annealing is only as good as the move set. Pair it with polish, B-star, or sequence-pair edits—plus soft sizing and macros—in the labs that follow.

- Penalty for illegal solutions
- Golden cost $\ll$ bad cost
- Representation defines neighbors

```
starter: improve to golden
```

SA needs a representation

Browser lab track

- Open simulated-annealing-fp
- Show bad, note cost at least one thousand
- Show golden or Improve, cost drops below one thousand and legality becomes true

Implement track

- Implement cost with an illegality penalty, plus deadspace and HPWL terms
- Assert cost(golden) is less than cost(bad), and saSwap only exchanges coordinates

Pitfalls

- Accepting illegal states without penalty

Your turn

- Demonstrate one improve step from bad to golden
- Next: soft module A reshaped from three by two to two by three